Class 12 Mathematics

Chapter 11: Three Dimensional Geometry

School Worksheet: step-wise solutions

20 questions **51** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 MCQ — 1 mark

The direction cosines of a line equally inclined to the coordinate axes are:

(A) $\pm 1, \pm 1, \pm 1$

(B) $\pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}$

(C) $\pm\frac{1}{3}, \pm\frac{1}{3}, \pm\frac{1}{3}$

(D) $\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}}$

Formula / idea: Direction cosines always satisfy $l^2 + m^2 + n^2 = 1$.

1 Equal angles with the axes give equal direction cosines.
$l = m = n$

2 Use $l^2 + m^2 + n^2 = 1$. ✓ 1
$3l^2 = 1 \Rightarrow l = \pm\frac{1}{\sqrt{3}}$

3 So each direction cosine is $\pm\frac{1}{\sqrt{3}}$, which is option (B).

Answer **(B)** $\pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}$

Watch out: Picking $\pm\frac{1}{3}$ by forgetting the square root in $l^2 = \frac{1}{3}$.

2 MCQ 1 mark

If a line makes angles $90°$, $135°$ and $45°$ with the x, y and z axes respectively, its direction cosines are:

(A) $0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

(B) $0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

(C) $1, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

(D) $0, -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}$

Formula / idea: If a line makes angles α, β, γ with the axes, its DCs are $\cos\alpha, \cos\beta, \cos\gamma$.

1 Take the cosine of each angle.
$l = \cos 90°,\ m = \cos 135°,\ n = \cos 45°$

2 Evaluate each cosine. ✓ 1
$l = 0,\ m = -\frac{1}{\sqrt{2}},\ n = \frac{1}{\sqrt{2}}$
$135°$ is in the second quadrant, so cosine is negative.

3 Check: $0 + \frac{1}{2} + \frac{1}{2} = 1$. So the answer is option (B).

Answer **(B)** $0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

Watch out: Writing $\cos 135° = +\frac{1}{\sqrt{2}}$ and choosing (A).

3 MCQ 1 mark

The Cartesian equation of the line through $(5, -2, 4)$ and parallel to $3\hat{i} + 2\hat{j} - 8\hat{k}$ is:

(A) $\frac{x+5}{3} = \frac{y-2}{2} = \frac{z+4}{-8}$

(B) $\frac{x-5}{3} = \frac{y+2}{2} = \frac{z-4}{-8}$

(C) $\frac{x-3}{5} = \frac{y-2}{-2} = \frac{z+8}{4}$

(D) $\frac{x-5}{3} = \frac{y+2}{2} = \frac{z-4}{8}$

Formula / idea: Line through (x_1, y_1, z_1) with DRs a, b, c: $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$.

1 Point $(5, -2, 4)$; DRs from the vector are $3, 2, -8$.
$x_1 = 5,\ y_1 = -2,\ z_1 = 4; \quad a = 3,\ b = 2,\ c = -8$

2 Substitute in the Cartesian form. ✓ 1
$$\frac{x-5}{3}=\frac{y+2}{2}=\frac{z-4}{-8}$$

3 This matches option (B).

Answer **(B)** $\frac{x-5}{3}=\frac{y+2}{2}=\frac{z-4}{-8}$

Watch out: Dropping the minus sign of -8 and choosing (D).

4 MCQ 1 mark

The angle between the lines $\frac{x-2}{2}=\frac{y-1}{5}=\frac{z+3}{-3}$ and $\frac{x+2}{-1}=\frac{y-4}{8}=\frac{z-5}{4}$ is:

(A) $\cos^{-1}\left(\frac{26}{9\sqrt{38}}\right)$

(B) $\cos^{-1}\left(\frac{26}{9}\right)$

(C) $\cos^{-1}\left(\frac{13}{9\sqrt{38}}\right)$

(D) $\frac{\pi}{2}$

Formula / idea: $\cos\theta=\frac{|a_1a_2+b_1b_2+c_1c_2|}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}$

1 Read the direction ratios of both lines.
$(2,5,-3)$ and $(-1,8,4)$

2 Find the dot product of the DRs.
$2(-1)+5(8)+(-3)(4)=-2+40-12=26$

3 Find both magnitudes.
$\sqrt{4+25+9}=\sqrt{38}, \quad \sqrt{1+64+16}=9$

4 Put into the formula. ✓ 1
$$\cos\theta=\frac{26}{9\sqrt{38}}\Rightarrow\theta=\cos^{-1}\left(\frac{26}{9\sqrt{38}}\right)$$

5 This is option (A).

Answer **(A)** $\cos^{-1}\left(\frac{26}{9\sqrt{38}}\right)$

Watch out: Taking $\sqrt{81}$ as 81, or halving 26 to get option (C).

5 MCQ 1 mark

The value of p for which the lines $\frac{1-x}{3} = \frac{7y-14}{2p} = \frac{z-3}{2}$ and $\frac{7-7x}{3p} = \frac{y-5}{1} = \frac{6-z}{5}$ are perpendicular is:

(A) $\frac{70}{11}$

(B) $\frac{11}{70}$

(C) $-\frac{70}{11}$

(D) 7

Formula / idea: Make the coefficients of x, y, z equal to 1 before reading DRs; perpendicular means $a_1a_2 + b_1b_2 + c_1c_2 = 0$.

1 Rewrite line 1 in standard form.
$\frac{x-1}{-3} = \frac{y-2}{\frac{2p}{7}} = \frac{z-3}{2}$

2 Rewrite line 2 in standard form.
$\frac{x-1}{-\frac{3p}{7}} = \frac{y-5}{1} = \frac{z-6}{-5}$

3 Apply the perpendicular condition to DRs $\left(-3, \frac{2p}{7}, 2\right)$ and $\left(-\frac{3p}{7}, 1, -5\right)$.
$\frac{9p}{7} + \frac{2p}{7} - 10 = 0$

4 Solve for p. ✓ 1
$\frac{11p}{7} = 10 \Rightarrow p = \frac{70}{11}$

5 This is option (A).

Answer **(A)** $\frac{70}{11}$

Watch out: Reading DRs as $3, 2p, 2$ straight from $\frac{1-x}{3}$ and $\frac{7y-14}{2p}$ without making coefficients 1.

6 **Assertion-Reason** 1 mark

Assertion (A): The line $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ passes through the origin. **Reason (R):** Any line of the form $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$ passes through the origin, since $(0, 0, 0)$ satisfies it.

(A) **Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).**

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: $\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$ passes through (x_1, y_1, z_1).

1	Put $(0, 0, 0)$ in the line of A. $\frac{0}{1} = \frac{0}{2} = \frac{0}{3} = 0$ All three ratios are equal, so the point lies on the line.	
2	So **A is true**.	
3	In $\frac{x - 0}{a} = \frac{y - 0}{b} = \frac{z - 0}{c}$ the fixed point is $(0, 0, 0)$. So **R is true**.	
4	The line in A is R with $a = 1, b = 2, c = 3$. So **R explains A**.	✓ 1
5	Hence the correct option is (A).	

Answer **(A) Both A and R are true and R is the correct explanation of A.**

Watch out: Choosing (B) because R is a general statement; here A is exactly a special case of R.

7 **Short answer** 2 marks

Find the vector and Cartesian equations of the line passing through $A(1, 2, -1)$ and $B(2, 1, 1)$.

Formula / idea: Line through two points: $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$.

1	Write the position vectors of A and B. $\vec{a} = \hat{i} + 2\hat{j} - \hat{k}, \quad \vec{b} = 2\hat{i} + \hat{j} + \hat{k}$	
2	Find the direction $\vec{b} - \vec{a}$. $\vec{b} - \vec{a} = (2 - 1)\hat{i} + (1 - 2)\hat{j} + (1 + 1)\hat{k} = \hat{i} - \hat{j} + 2\hat{k}$	✓ ½
3	Vector equation: $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$. $\vec{r} = (\hat{i} + 2\hat{j} - \hat{k}) + \lambda(\hat{i} - \hat{j} + 2\hat{k})$	✓ ½

4	Cartesian form uses point $(1, 2, -1)$ and DRs $1, -1, 2$. $\frac{x-1}{1} = \frac{y-2}{-1} = \frac{z+1}{2}$	✓ 1
5	These are the required vector and Cartesian equations.	

Answer $\vec{r} = (\hat{i} + 2\hat{j} - \hat{k}) + \lambda(\hat{i} - \hat{j} + 2\hat{k})$; $\frac{x-1}{1} = \frac{y-2}{-1} = \frac{z+1}{2}$

Watch out: Writing $\vec{r} = \vec{a} + \lambda\vec{b}$, using B's position vector as the direction.

Level 2

16 marks

8 **Short answer** 2 marks

Show that the lines $\frac{x-5}{7} = \frac{y+2}{-5} = \frac{z}{1}$ and $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ are perpendicular to each other.

Formula / idea: Lines are perpendicular if $a_1a_2 + b_1b_2 + c_1c_2 = 0$.

1	Read the direction ratios of both lines. $a_1, b_1, c_1 = 7, -5, 1; \quad a_2, b_2, c_2 = 1, 2, 3$	✓ ½
2	State the condition for perpendicular lines. $a_1a_2 + b_1b_2 + c_1c_2 = 0$	✓ ½
3	Substitute the DRs. $7(1) + (-5)(2) + 1(3) = 7 - 10 + 3 = 0$	✓ ½
4	The condition holds. Hence the two lines are perpendicular.	✓ ½

Answer $7(1) + (-5)(2) + 1(3) = 0$**, so the lines are perpendicular.**

Watch out: Using the points $(5, -2, 0)$ and $(0, 0, 0)$ instead of the denominators.

9 **Short answer** 3 marks

Find the shortest distance between the lines $\vec{r} = (\hat{i} + 2\hat{j} + \hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$ and $\vec{r} = (2\hat{i} - \hat{j} - \hat{k}) + \mu(2\hat{i} + \hat{j} + 2\hat{k})$.

Formula / idea: $d = \left| \frac{(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1)}{|\vec{b}_1 \times \vec{b}_2|} \right|$

1	Compare with $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu\vec{b}_2$. $\vec{a}_1 = \hat{i} + 2\hat{j} + \hat{k}, \quad \vec{b}_1 = \hat{i} - \hat{j} + \hat{k}$ $\vec{a}_2 = 2\hat{i} - \hat{j} - \hat{k}, \quad \vec{b}_2 = 2\hat{i} + \hat{j} + 2\hat{k}$	

2 Find $\vec{a}_2 - \vec{a}_1$. ✓ ½

$\vec{a}_2 - \vec{a}_1 = \hat{i} - 3\hat{j} - 2\hat{k}$

3 Find $\vec{b}_1 \times \vec{b}_2$. ✓ 1

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 1 \\ 2 & 1 & 2 \end{vmatrix} = \hat{i}(-2-1) - \hat{j}(2-2) + \hat{k}(1+2) = -3\hat{i} + 3\hat{k}$$

4 Find its magnitude. ✓ ½

$|\vec{b}_1 \times \vec{b}_2| = \sqrt{9+0+9} = 3\sqrt{2}$

5 Find the dot product with $\vec{a}_2 - \vec{a}_1$. ✓ ½

$(-3)(1) + 0(-3) + 3(-2) = -9$

6 Put into the formula. ✓ ½

$$d = \left|\frac{-9}{3\sqrt{2}}\right| = \frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2} \text{ units}$$

Answer **Shortest distance** $= \frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$ **units**

Watch out: Missing the minus sign on the middle ($\hat{j}$) term of the determinant.

10 Short answer — 3 marks

Find the distance between the parallel lines $\vec{r} = \hat{i} + 2\hat{j} - 4\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$ and $\vec{r} = 3\hat{i} + 3\hat{j} - 5\hat{k} + \mu(2\hat{i} + 3\hat{j} + 6\hat{k})$.

Formula / idea: Parallel lines: $d = \left|\dfrac{\vec{b} \times (\vec{a}_2 - \vec{a}_1)}{|\vec{b}|}\right|$

1 Read the common direction and the two points.

$\vec{b} = 2\hat{i} + 3\hat{j} + 6\hat{k}, \quad \vec{a}_1 = \hat{i} + 2\hat{j} - 4\hat{k}, \quad \vec{a}_2 = 3\hat{i} + 3\hat{j} - 5\hat{k}$

2 State the formula for parallel lines. ✓ ½

$$d = \left|\frac{\vec{b} \times (\vec{a}_2 - \vec{a}_1)}{|\vec{b}|}\right|$$

3 Find $\vec{a}_2 - \vec{a}_1$. ✓ ½

$\vec{a}_2 - \vec{a}_1 = 2\hat{i} + \hat{j} - \hat{k}$

4 Find $\vec{b} \times (\vec{a}_2 - \vec{a}_1)$. ✓ 1

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 6 \\ 2 & 1 & -1 \end{vmatrix} = \hat{i}(-3-6) - \hat{j}(-2-12) + \hat{k}(2-6) = -9\hat{i} + 14\hat{j} - 4\hat{k}$$

5 Find both magnitudes. ✓ ½

$\sqrt{81+196+16} = \sqrt{293}, \quad |\vec{b}| = \sqrt{4+9+36} = 7$

6 Put into the formula.

$d = \dfrac{\sqrt{293}}{7}$ units

✓ ½

Answer **Distance** $= \dfrac{\sqrt{293}}{7}$ **units**

Watch out: Using the skew-lines formula here; $\vec{b}_1 \times \vec{b}_2 = \vec{0}$ for parallel lines.

11 Short answer — 2 marks

Find the angle between the lines $\vec{r} = 3\hat{i} + 2\hat{j} - 4\hat{k} + \lambda(\hat{i} + 2\hat{j} + 2\hat{k})$ and $\vec{r} = 5\hat{i} - 2\hat{j} + \mu(3\hat{i} + 2\hat{j} + 6\hat{k})$.

Formula / idea: $\cos\theta = \dfrac{|\vec{b}_1 \cdot \vec{b}_2|}{|\vec{b}_1|\,|\vec{b}_2|}$

1 Read the direction vectors.

$\vec{b}_1 = \hat{i} + 2\hat{j} + 2\hat{k}, \quad \vec{b}_2 = 3\hat{i} + 2\hat{j} + 6\hat{k}$

2 State the formula.

$\cos\theta = \dfrac{|\vec{b}_1 \cdot \vec{b}_2|}{|\vec{b}_1|\,|\vec{b}_2|}$

✓ ½

3 Find the dot product.

$\vec{b}_1 \cdot \vec{b}_2 = 3 + 4 + 12 = 19$

✓ ½

4 Find the magnitudes.

$|\vec{b}_1| = \sqrt{1+4+4} = 3, \quad |\vec{b}_2| = \sqrt{9+4+36} = 7$

✓ ½

5 Substitute and write the angle.

$\cos\theta = \dfrac{19}{3 \times 7} = \dfrac{19}{21} \Rightarrow \theta = \cos^{-1}\left(\dfrac{19}{21}\right)$

✓ ½

Answer $\theta = \cos^{-1}\left(\dfrac{19}{21}\right)$

Watch out: Using the position vectors $3\hat{i} + 2\hat{j} - 4\hat{k}$ and $5\hat{i} - 2\hat{j}$ instead of $\vec{b}_1, \vec{b}_2$.

12 Short answer — 2 marks

Find k so that the lines $\dfrac{x-1}{-3} = \dfrac{y-2}{2k} = \dfrac{z-3}{2}$ and $\dfrac{x-1}{3k} = \dfrac{y-1}{1} = \dfrac{z-6}{-5}$ are perpendicular.

Formula / idea: Lines are perpendicular if $a_1a_2 + b_1b_2 + c_1c_2 = 0$.

1 Read the direction ratios.

$(-3, 2k, 2)$ and $(3k, 1, -5)$

✓ ½

2 Apply the perpendicular condition. ✓ ½
$(-3)(3k) + (2k)(1) + (2)(-5) = 0$

3 Simplify. ✓ ½
$-9k + 2k - 10 = 0 \Rightarrow -7k = 10$

4 Hence the required value is ✓ ½
$k = -\dfrac{10}{7}$

Answer $k = -\dfrac{10}{7}$

Watch out: Sign slip: writing $-7k = -10$ and getting $k = \dfrac{10}{7}$.

13 Short answer — 2 marks

Find the direction cosines of the line whose Cartesian equations are $5x - 3 = 15y + 7 = 3 - 10z$.

Formula / idea: Write the line as $\dfrac{x - x_1}{a} = \dfrac{y - y_1}{b} = \dfrac{z - z_1}{c}$; then DCs $= \dfrac{a}{\sqrt{a^2 + b^2 + c^2}}$, etc.

1 Take out the coefficient of x, y, z in each part. ✓ ½
$5\left(x - \dfrac{3}{5}\right) = 15\left(y + \dfrac{7}{15}\right) = -10\left(z - \dfrac{3}{10}\right)$

2 Divide every part by 30 (LCM of $5, 15, 10$). ✓ ½
$\dfrac{x - \frac{3}{5}}{6} = \dfrac{y + \frac{7}{15}}{2} = \dfrac{z - \frac{3}{10}}{-3}$

3 Read the DRs and find their magnitude. ✓ ½
$a, b, c = 6, 2, -3$; $\quad \sqrt{36 + 4 + 9} = 7$

4 Divide each DR by 7 to get the DCs. ✓ ½
$l = \dfrac{6}{7},\ m = \dfrac{2}{7},\ n = -\dfrac{3}{7}$

Answer $\dfrac{6}{7}, \dfrac{2}{7}, -\dfrac{3}{7}$

Watch out: Taking DRs as $5, 15, -10$; the coefficients of x, y, z must be 1 first.

14 **Short answer** 2 marks

Find the Cartesian equation of the line through $(-2, 4, -5)$ and parallel to $\frac{x+3}{3} = \frac{y-4}{5} = \frac{z+8}{6}$.

Formula / idea: Parallel lines have the same direction ratios.

1	Read the DRs of the given line. $a, b, c = 3, 5, 6$	✓ ½
2	The required line is parallel, so it has the same DRs $3, 5, 6$.	✓ ½
3	Use the line through (x_1, y_1, z_1) with DRs a, b, c. $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$	✓ ½
4	Put $(x_1, y_1, z_1) = (-2, 4, -5)$. $\frac{x+2}{3} = \frac{y-4}{5} = \frac{z+5}{6}$	✓ ½

Answer $\frac{x+2}{3} = \frac{y-4}{5} = \frac{z+5}{6}$

Watch out: Keeping the point $(-3, 4, -8)$ of the given line instead of $(-2, 4, -5)$.

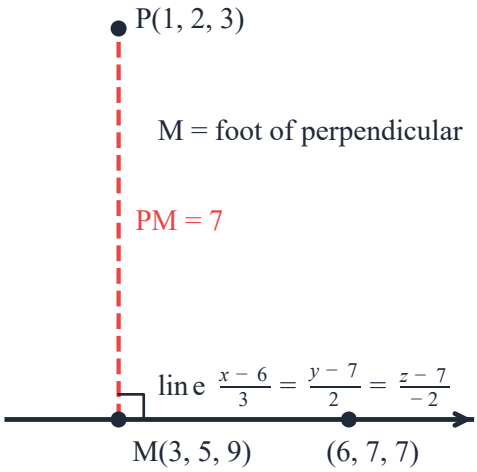
Level 3

27 marks

15 **Long answer** 5 marks

Find the coordinates of the foot of the perpendicular drawn from $P(1, 2, 3)$ to the line $\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}$. Also find the length of the perpendicular.

Formula / idea: Take a general point M on the line; $PM \perp$ line gives λ.



1	Let M be the foot. Put the line equal to λ; M is a general point. $M(3\lambda + 6,\ 2\lambda + 7,\ -2\lambda + 7)$	✓ 1
2	DRs of PM = coordinates of M minus P. $(3\lambda + 5,\ 2\lambda + 5,\ -2\lambda + 4)$	✓ 1
3	$PM \perp$ line, whose DRs are $3, 2, -2$. $3(3\lambda + 5) + 2(2\lambda + 5) - 2(-2\lambda + 4) = 0$	✓ ½
4	Simplify and solve. $9\lambda + 15 + 4\lambda + 10 + 4\lambda - 8 = 0 \Rightarrow 17\lambda + 17 = 0 \Rightarrow \lambda = -1$	✓ ½
5	Put $\lambda = -1$ in M. $M(-3 + 6,\ -2 + 7,\ 2 + 7) = M(3, 5, 9)$	✓ 1
6	Length PM by the distance formula. $PM = \sqrt{(3-1)^2 + (5-2)^2 + (9-3)^2} = \sqrt{4 + 9 + 36} = \sqrt{49}$	✓ ½
7	Hence the foot is $(3, 5, 9)$ and the length of the perpendicular is $PM = 7$ units	✓ ½

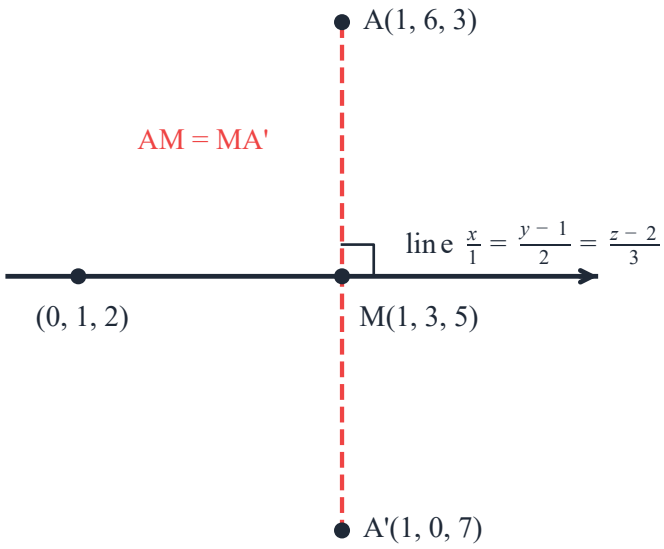
Answer **Foot of perpendicular $M(3, 5, 9)$; length of perpendicular $= 7$ units**

Watch out: Writing $-2(-2\lambda + 4)$ as $-4\lambda - 8$; it is $+4\lambda - 8$.

16 **Long answer** 5 marks

Find the image A' of the point $A(1, 6, 3)$ in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$. Also, find the equation of the line joining A and A'.

Formula / idea: Foot M of the perpendicular from A is the mid-point of A and its image A'.



1	Put the line equal to λ. Let M, the foot, be a general point. $M(\lambda,\ 2\lambda+1,\ 3\lambda+2)$	✓ ½
2	DRs of AM. $(\lambda-1,\ 2\lambda-5,\ 3\lambda-1)$	✓ ½
3	$AM \perp$ line, whose DRs are $1, 2, 3$. $1(\lambda-1)+2(2\lambda-5)+3(3\lambda-1)=0$	✓ ½
4	Simplify and solve. $14\lambda-14=0 \Rightarrow \lambda=1$	✓ ½
5	Put $\lambda=1$: foot of perpendicular. $M(1,3,5)$	✓ ½
6	Let $A'(\alpha,\beta,\gamma)$. M is the mid-point of AA'. $\frac{1+\alpha}{2}=1,\quad \frac{6+\beta}{2}=3,\quad \frac{3+\gamma}{2}=5$	✓ ½
7	Solve each equation. $\alpha=1,\ \beta=0,\ \gamma=7 \Rightarrow A'(1,0,7)$	✓ ½
8	DRs of AA'. $(1-1,\ 0-6,\ 7-3)=(0,-6,4)\ \propto\ (0,-3,2)$	✓ ½

9 Line through $A(1,6,3)$ along $-3\hat{j}+2\hat{k}$, in vector form. ✓ ½
$\vec{r}=(\hat{i}+6\hat{j}+3\hat{k})+\mu(-3\hat{j}+2\hat{k})$

10 Cartesian form ($\frac{x-1}{0}$ only means $x=1$ on this line). ✓ ½
$\frac{x-1}{0}=\frac{y-6}{-3}=\frac{z-3}{2}$

Answer **Image $A'(1,0,7)$; line AA': $\vec{r}=(\hat{i}+6\hat{j}+3\hat{k})+\mu(-3\hat{j}+2\hat{k})$, i.e. $\frac{x-1}{0}=\frac{y-6}{-3}=\frac{z-3}{2}$**

Watch out: Giving the foot $M(1,3,5)$ as the image; the image is twice as far, $A'=2M-A$.

17 Long answer — 5 marks

Show that the lines $\frac{x+1}{3}=\frac{y+3}{5}=\frac{z+5}{7}$ and $\frac{x-2}{1}=\frac{y-4}{3}=\frac{z-6}{5}$ intersect. Find their point of intersection.

Formula / idea: Lines intersect if λ, μ found from two coordinates also satisfy the third.

1 General point on line 1 (put it equal to λ). ✓ ½
$(3\lambda-1,\ 5\lambda-3,\ 7\lambda-5)$

2 General point on line 2 (put it equal to μ). ✓ ½
$(\mu+2,\ 3\mu+4,\ 5\mu+6)$

3 Equate the x-coordinates. ✓ ½
$3\lambda-1=\mu+2\Rightarrow 3\lambda-\mu=3 \quad \ldots(1)$

4 Equate the y-coordinates. ✓ ½
$5\lambda-3=3\mu+4\Rightarrow 5\lambda-3\mu=7 \quad \ldots(2)$

5 Multiply (1) by 3 and subtract (2). ✓ 1
$9\lambda-3\mu-(5\lambda-3\mu)=9-7$
$4\lambda=2\Rightarrow\lambda=\frac{1}{2}, \quad \mu=3\lambda-3=-\frac{3}{2}$

6 Check the z-coordinates with these values. ✓ ½
$7\lambda-5=\frac{7}{2}-5=-\frac{3}{2}, \qquad 5\mu+6=-\frac{15}{2}+6=-\frac{3}{2}$

7 The z-coordinates are equal, so the lines **intersect**. ✓ ½

8 Put $\lambda=\frac{1}{2}$ in the point of line 1. ✓ 1
$\left(\frac{3}{2}-1,\ \frac{5}{2}-3,\ -\frac{3}{2}\right)=\left(\frac{1}{2},-\frac{1}{2},-\frac{3}{2}\right)$

Answer **The lines intersect at $\left(\frac{1}{2},-\frac{1}{2},-\frac{3}{2}\right)$.**

Watch out: Not checking the third (z) equation; without it you have not shown that the lines intersect.

18 **Long answer** 5 marks

Find a point P on the line $\frac{x+5}{1} = \frac{y+3}{4} = \frac{z-6}{-9}$ such that its distance from the point $Q(2, 4, -1)$ is 7 units. Also, find the equation of the line joining P and Q.

Formula / idea: Take a general point on the line and use the distance formula $PQ^2 = 49$.

1	Put the line equal to λ. A general point on it is $P(\lambda - 5,\ 4\lambda - 3,\ -9\lambda + 6)$	✓ 1
2	Use $PQ = 7$, so $PQ^2 = 49$. $(2 - \lambda + 5)^2 + (4 - 4\lambda + 3)^2 + (-1 + 9\lambda - 6)^2 = 49$ $(7 - \lambda)^2 + (7 - 4\lambda)^2 + (9\lambda - 7)^2 = 49$	✓ 1
3	Expand and add. $98\lambda^2 - 196\lambda + 147 = 49$	✓ ½
4	Simplify and solve. $98\lambda^2 - 196\lambda + 98 = 0 \Rightarrow (\lambda - 1)^2 = 0 \Rightarrow \lambda = 1$	✓ ½
5	Put $\lambda = 1$ in P. $P(1 - 5,\ 4 - 3,\ -9 + 6) = P(-4, 1, -3)$	✓ 1
6	DRs of PQ. $(2 + 4,\ 4 - 1,\ -1 + 3) = (6, 3, 2)$	✓ ½
7	Line through $Q(2, 4, -1)$ with DRs $6, 3, 2$. $\frac{x-2}{6} = \frac{y-4}{3} = \frac{z+1}{2}$	✓ ½

Answer $P(-4, 1, -3)$; **line** PQ: $\frac{x-2}{6} = \frac{y-4}{3} = \frac{z+1}{2}$

Watch out: Squaring errors: $(9\lambda - 7)^2 = 81\lambda^2 - 126\lambda + 49$, not $81\lambda^2 + 49$.

19 **Short answer** 3 marks

Find the vector equation of the line passing through $(1, 2, -4)$ and perpendicular to both the lines $\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}$ and $\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$.

Formula / idea: A line perpendicular to both $\vec{b}_1$ and $\vec{b}_2$ is parallel to $\vec{b}_1 \times \vec{b}_2$.

1	Write the direction vectors of the two lines. $\vec{b}_1 = 3\hat{i} - 16\hat{j} + 7\hat{k}, \quad \vec{b}_2 = 3\hat{i} + 8\hat{j} - 5\hat{k}$	✓ ½
2	The required line is perpendicular to both, so it is parallel to $\vec{b}_1 \times \vec{b}_2$.	✓ ½

3 Find $\vec{b}_1 \times \vec{b}_2$.
$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -16 & 7 \\ 3 & 8 & -5 \end{vmatrix} = \hat{i}(80-56) - \hat{j}(-15-21) + \hat{k}(24+48)$$ ✓ ½

4 Simplify and take out the common factor 12.
$= 24\hat{i} + 36\hat{j} + 72\hat{k} = 12(2\hat{i} + 3\hat{j} + 6\hat{k})$ ✓ ½

5 So take $\vec{b} = 2\hat{i} + 3\hat{j} + 6\hat{k}$ and $\vec{a} = \hat{i} + 2\hat{j} - 4\hat{k}$.

6 Required line $\vec{r} = \vec{a} + \lambda\vec{b}$.
$\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$ ✓ 1

Answer $\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$

Watch out: Middle term sign: $-\hat{j}(-15-21) = +36\hat{j}$, not $-36\hat{j}$.

20 Case study — 4 marks

Case Study: Two aircraft fly along straight paths given by $L_1 : \vec{r} = (\hat{i} + \hat{j}) + \lambda(2\hat{i} - \hat{j} + \hat{k})$ and $L_2 : \vec{r} = (2\hat{i} + \hat{j} - \hat{k}) + \mu(3\hat{i} - 5\hat{j} + 2\hat{k})$. Air-traffic control needs to know how close the paths come.

(i) Write the direction ratios of L_1 and L_2. — 1 mark

(ii) Are the paths parallel? Justify. — 1 mark

(iii) Find the shortest distance between the two paths. — 2 marks

Formula / idea: $d = \left| \dfrac{(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1)}{|\vec{b}_1 \times \vec{b}_2|} \right|$

(i) — 1 mark

1 DRs of L_1 are the coefficients in $\vec{b}_1 = 2\hat{i} - \hat{j} + \hat{k}$.
$2, -1, 1$ ✓ ½

2 DRs of L_2 are the coefficients in $\vec{b}_2 = 3\hat{i} - 5\hat{j} + 2\hat{k}$.
$3, -5, 2$ ✓ ½

Answer (i) L_1: $2, -1, 1$; L_2: $3, -5, 2$

(ii) — 1 mark

1 Compare the ratios of the DRs.
$\dfrac{2}{3} \neq \dfrac{-1}{-5} = \dfrac{1}{5}$ ✓ ½

2 DRs are not proportional, so the paths are **not parallel**. ✓ ½

Answer (ii) **No, the paths are not parallel, since** $\dfrac{2}{3} \neq \dfrac{-1}{-5}$.

(iii) 2 marks

1 Find $\vec{a}_2 - \vec{a}_1$ with $\vec{a}_1 = \hat{i} + \hat{j}$, $\vec{a}_2 = 2\hat{i} + \hat{j} - \hat{k}$. ✓ ½

$\vec{a}_2 - \vec{a}_1 = \hat{i} + 0\hat{j} - \hat{k}$

2 Find $\vec{b}_1 \times \vec{b}_2$. ✓ ½

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 1 \\ 3 & -5 & 2 \end{vmatrix} = \hat{i}(-2+5) - \hat{j}(4-3) + \hat{k}(-10+3) = 3\hat{i} - \hat{j} - 7\hat{k}$$

3 Find its magnitude and the dot product. ✓ ½

$|\vec{b}_1 \times \vec{b}_2| = \sqrt{9+1+49} = \sqrt{59}; \quad (3)(1) + (-1)(0) + (-7)(-1) = 10$

4 Put into the shortest distance formula. ✓ ½

$$d = \left| \frac{10}{\sqrt{59}} \right| = \frac{10}{\sqrt{59}} \text{ units}$$

Answer (iii) **Shortest distance** $= \dfrac{10}{\sqrt{59}} = \dfrac{10\sqrt{59}}{59}$ **units**

Watch out: Taking $\vec{a}_1 = \hat{i} + \hat{j} + \hat{k}$; L_1 has no $\hat{k}$ term, so $\vec{a}_1 = \hat{i} + \hat{j}$.